COSC265 Relational Database Systems – Course Outline

Dr Tanja Mitrovic – Course Coordinator
June 2026

Introduction

This course covers data models and database systems and has been designed to enable students to understand the theoretical basis of databases and apply that knowledge to develop and use relational databases. The lectures are organized around the database design lifecycle, while in the labs, students develop a database from given requirements. Students who achieve a good grade in the course should be able to:

- Understand the role of database systems in software applications
- Understand the database design lifecycle
- Design conceptual database schemas starting from requirements
- Use SQL to implement, query, populate, update, and manage databases
- Normalize relational databases
- Design physical schemas to achieve good performance
- Understand and use database management systems (DBMS)

Lectures, Labs, and Lecturers

There will be three lectures per week (times to be announced), delivered by Prof Tanja Mitrovic (email: tanja.mitrovic@canterbury.ac.nz) and Associate Professor Miguel Morales (miguel.morales@canterbury.ac.nz). Labs will be held in CSSE labs in the Jack Erskine building.

Please consult the CIS web page and your timetable for the latest information on the times and locations of lectures, labs, and tutorials; the CSSE Department is not responsible for scheduling them. More information about the course will be available on LEARN.

Assessment

Item	Weight	Date
Assignment 1	8%	07 August 2026, 5pm
Assignment 2	16%	21 August 2026, 5pm
Quizzes	6%	TBA
Lab Test	20%	01 October 2026, 6:30-8:30pm
Final exam (closed book, 2 hours)	50%	TBA

No assignments will be accepted after the drop-dead date (i.e., a week after the assignment is due). The penalty for the late submission of an assignment will be an *absolute* deduction of 15% of the maximum possible mark.

In order to pass a course, you must meet two requirements:

1. You must achieve an average grade of at least 50% over all assessment items.
2. You must achieve an average mark of at least 45% on invigilated assessment items.

If you satisfy both these criteria, your grade will be determined by the following University-wide scale for converting marks to grades: an average mark of 50% is sufficient for a C- grade, an average mark of 55% earns a C grade, 60% earns a C+ grade, and so forth. However, if you do not satisfy both the passing criteria, you will be given either a D or an E grade depending on your marks. Marks are sometimes scaled to achieve consistency between courses from year to year.

Required text

- Fundamentals of Database Systems (7th edition), Ramez Elmasri & Shamkant Navathe, Addison-Wesley, 2017. Previous editions of the same textbook are also suitable.
- The COSC265 lecture notes and all additional material will be available via Learn.

Important documents

COSC265 handouts will be available on LEARN. Notices about the course will be posted to the course forum on LEARN. CSSE students will also be made members of a class called "CSSE Notices", where general notices applicable to all classes (such as information about building access or job opportunities) will be posted.

Several important documents on departmental regulations, policies, and guidelines are available at All Departmental Policies. We expect students to be familiar with these.

Tentative lecture/lab outline

The course consists of 32 lectures, which means there will be some free lecture slots throughout the semester.

Week	Lecture topic	Lab/Tutorial
1	Database systems, database design lifecycle	No labs
2	Conceptual database design	Tutorial 1: Conceptual Design
3	Relational data model	EER-Tutor
4	Relational algebra & SQL DDL	Tutorial 2: EER Modeling
5	SQL Queries	Lab 1: DDL
6	SQL Queries & SQL DML	Lab 2: Basic DML
Break		
7	SQL DML	SQL-Tutor
8	SQL triggers & Data normalization	Lab 3: Advanced DML
9	Data normalization	Lab 4: Constraints, triggers
10	Catalogs & Physical level	Lab Test
11	Transaction Processing	Tutorial 3: Normalization
12	Query optimization & Course review	Tutorial 4: Physical Level